

A real ETF, read as a climate risk book

The estimand

For the SPDR MSCI EMU UCITS ETF (ISIN IE00B910VR50), a real Eurozone equity portfolio published without an account: (a) the share of market value in climate-policy-relevant sectors (Battiston et al. 2017), and (b) the portfolio's weighted average carbon intensity (WACI), in tCO₂e per EUR million of gross value added, with an interval reflecting proxy uncertainty.

This is a proxy, not a measurement

No company reports its own emissions here. Every holding gets its home country's national average intensity for one representative NACE activity standing in for its whole GICS sector. PCAF data-quality score 5 on every line -- this deck is the reason to read that score before the number.

The answer

15.3% of this EUR 353mn, 215-holding portfolio
(as of 25-Sep-2026) sits in climate-policy-relevant sectors.
WACI at 2024: 142 tCO₂e/EUR mn GVA, interval [139, 309]
from the spread of the same sector's intensity across EU member states --
right-skewed, because a few high-carbon member states are a real possibility
for any company sharing that sector code.

Method: a coarse, honest join

Data, all fetched by code

Holdings: SSGA's own daily holdings file for this fund, ssga.com, no key, no account. Intensity: Eurostat env_ac_ainah_r2 (GHG by NACE) over nama_10_a64 (gross value added by NACE), public API, all 27 EU member states. Physical overlay: modules/01's hazard table, already in this repository.

The join, and how coarse it is

The holdings file carries an MSCI GICS sector label (11 categories), not a NACE code. `src/portfolio_risk.py` maps each GICS sector to the single NACE Rev.2 code that best represents where that sector's value actually sits in this portfolio (e.g. Information Technology -> C26, computer & optical products, because ASML alone is 9% of the fund). Every company sharing a GICS label gets the same number regardless of its actual activity. `notes.md` has the reasoning sector by sector, and the sensitivity page shows what the coarseness costs.

A real data gap, not a bug

16.1% of portfolio value sits in a country/sector cell

Eurostat's national accounts don't break out at division level (Germany and Spain do not report NACE-division GVA for several manufacturing codes, likely statistical disclosure control). Those holdings fall back to the cross-country median for their NACE code, flagged in results/ rather than silently averaged in.

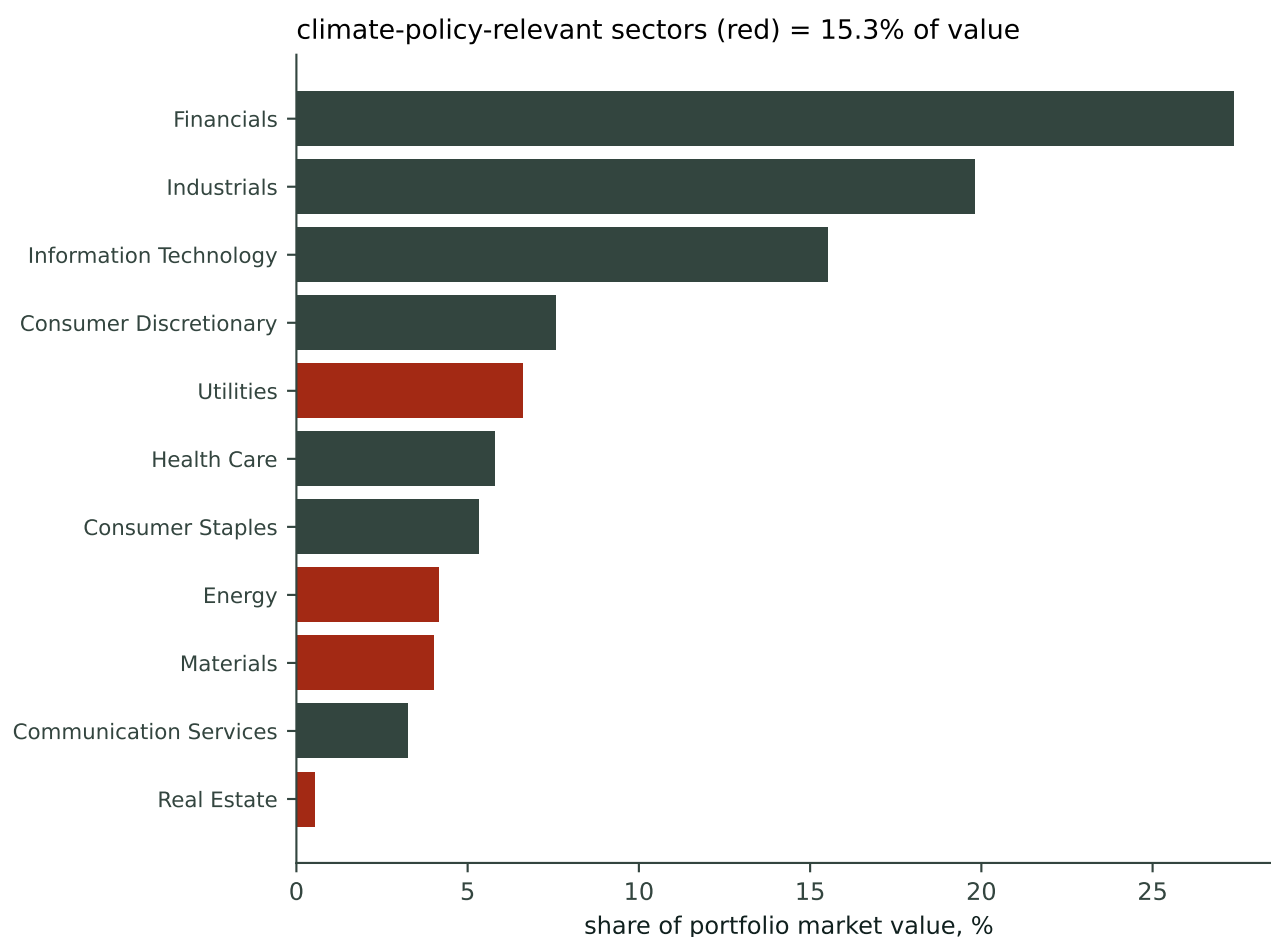
Result: where the exposure sits

Sector composition

Financials and Industrials dominate this fund by construction (it tracks large- and mid-cap Eurozone equities); climate-policy relevance is concentrated in four sectors that together are a minority of the book.

The four CPRS categories present

utilities: 6.6%, fossil_fuel: 4.2%, energy_intensive: 4.0%, housing: 0.5%.



Validation, and a physical overlay

Against the fund's own disclosure

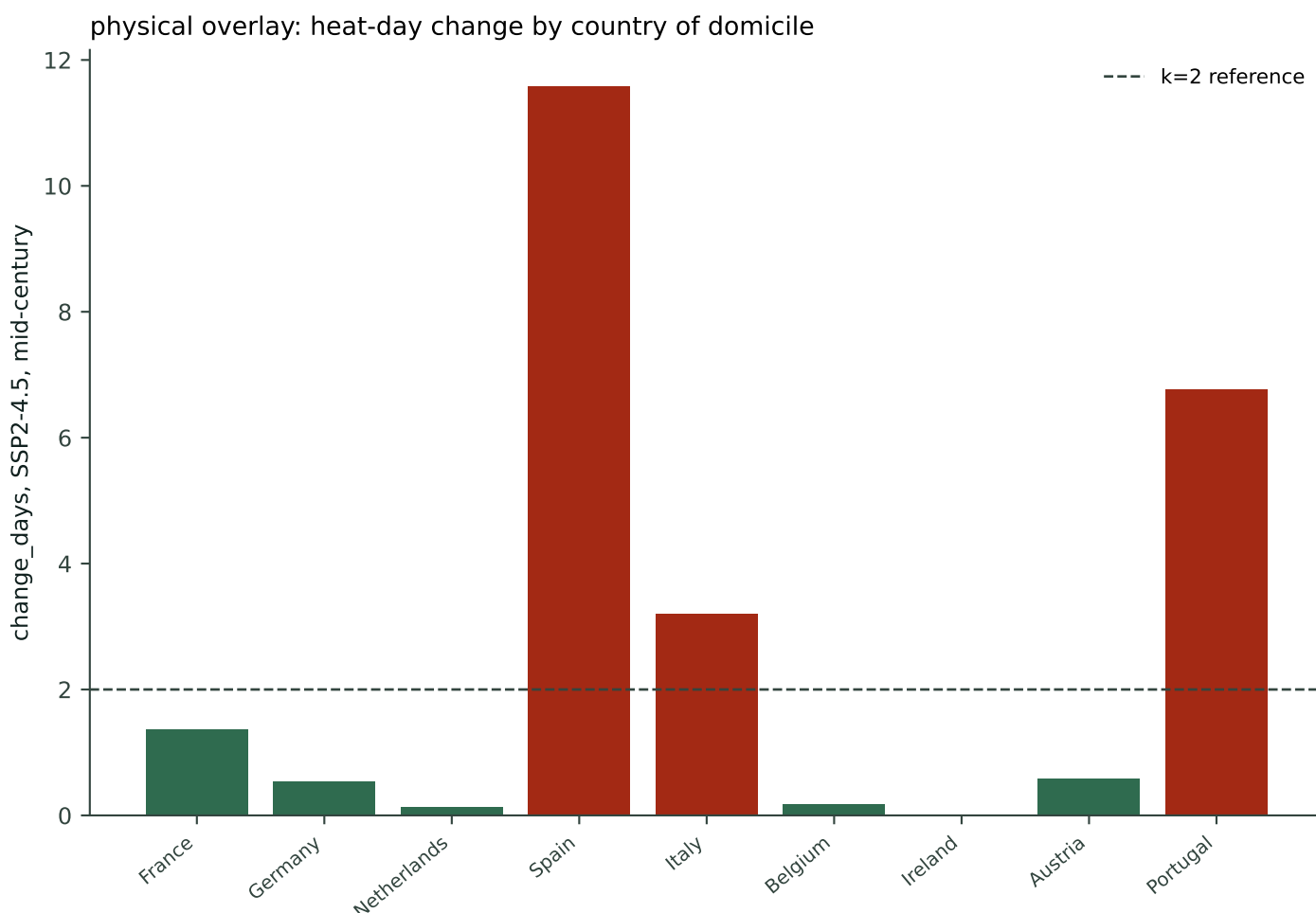
State Street's own per-ISIN sustainability report discloses an MSCI Weighted Average Carbon Intensity (Scope 1+2+3) of 1014.5 tCO₂e/\$M Sales -- fetched from ssga.com and read with a minimal stdlib PDF text extractor, because that figure exists nowhere else machine-readable for this fund.

Why it is not a like-for-like check

Different currency (USD sales vs EUR value added), different denominator (sales can be several times value added for the same firm), and MSCI's figure is over 90% Scope 3 -- supply-chain emissions this module's NACE-level, production-based proxy has no analogue for at all. Both point the same direction; neither confirms the other's number. Where no comparable metric exists, STANDARD.md says to say so rather than force a comparison -- this is that case.

Physical overlay, by country of domicile

Weighting module 01's SSP2-4.5 heat-day change by market value: mean 2.33 days [1.51, 3.31]. 22.7% of covered value (Finland is not in module 01's table and is excluded) sits in a country crossing the k=2 reference used in modules 01 and 03.



Sensitivity: mapping and vintage, not the model

The choice nobody can justify from first principles

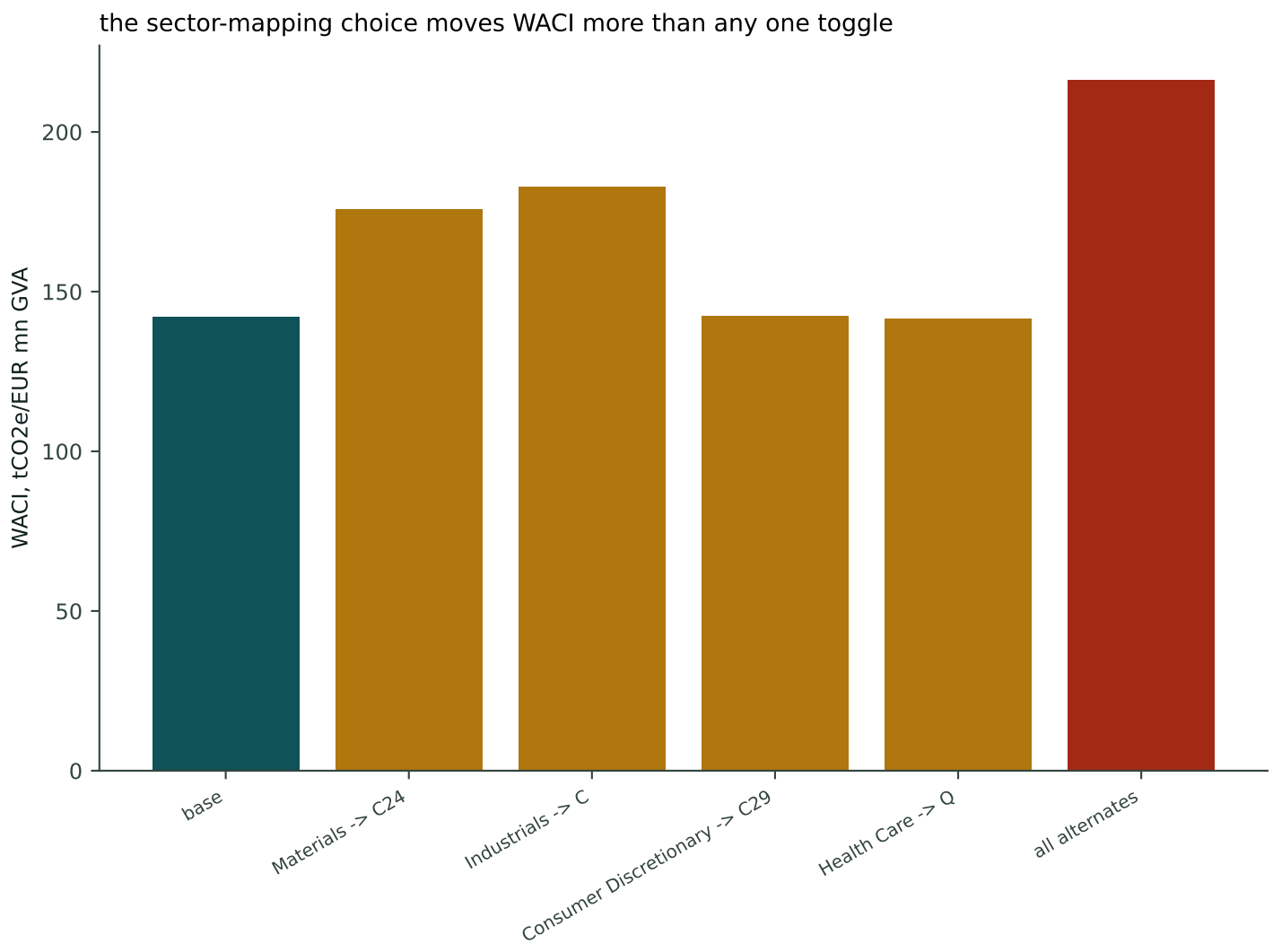
Which single NACE code stands in for a GICS sector. Re-running with the next most defensible reading for the four sectors where this portfolio holds a real mix (see notes.md) moves WACI from 142 (base) to 216 (all alternates) -- a 52% swing from a labelling choice, not from any data update.

A finding the number alone would hide

Reclassifying Consumer Discretionary from retail trade to motor-vehicle manufacture barely moves WACI (142 vs 142) but moves the CPRS share from 15.3% to 22.9%, because 'transportation' is a Battiston CPRS category and 'retail trade' is not. The two headline numbers are sensitive to different choices.

The reference year

Sweeping 2015-2024 (all 10 years Eurostat has): WACI falls from 409 to 142, monotonically, tracking the EU's own realised industrial decarbonisation and GVA growth -- not a projection, so no backtest applies; see the limits page.



Limits, and what I would build next

The assumption that breaks the conclusion

That a national-sector average is a reasonable stand-in for one company. It is not, for exactly the companies that matter most to the answer: ASML, Siemens and SAP are each far from their sector's national average by construction (they are the reason the sector has that average). The proxy is most defensible for the Financials-heavy middle of the book and least defensible for its largest, most distinctive single names -- the opposite of where a reader's attention goes.

The mapping is many-to-one by design

One NACE code per GICS sector cannot separate a chemicals company from a mining company inside 'Materials', or an airline from a machinery maker inside 'Industrials'. The sensitivity page shows this costs more than the year chosen or the interval width -- it is the first thing a skeptical reviewer should attack.

No backtest, and why not one here

STANDARD.md requires backtesting anything forward-looking. Nothing in this module is: the portfolio is a snapshot of today's holdings, the intensity is a realised national statistic, and the CPRS classification is a fixed taxonomy. The year sweep on the previous page is a robustness check on which vintage to use, not a forecast, so there is no future value to hold out and test against.

Next

Module 07 (PACTA alignment) and module 09 (financed emissions, PCAF) both need company-level activity data this module does not have. The natural next step for this module specifically is repeating it against a non-Eurozone, non-EU holding set (module 01's hazard table and Eurostat's NACE intensities both stop at the EU border), to see how much of this proxy's defensibility was borrowed from picking a portfolio that happens to sit inside the data's own coverage.